

CE 310 — Week 5 Weekly Quiz

Week: 5 | **Topic:** Normal Distribution — Concepts, Excel Functions, and Engineering Applications

Due: Closes **8:00 AM Tuesday, Sep 29** — the day before Exercise 5 and Homework 5 are due. Taken on D2L

Points: 15 questions $\times$ 1 pt each = 15 pts

Attempts: Two — the higher score counts

Time limit: 25 minutes

Instructions: Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

This quiz covers the Monday lecture and the Wednesday Excel tutorial — the Normal and lognormal distributions, NORM.DIST and NORM.INV, the empirical rule, and how ACI 318 acceptance criteria use them. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

Questions

Q1. Your concrete dataset has $\mu = 4,496$ psi and $\sigma = 442$ psi. A specification accepts cylinders whose strength falls between 3,800 and 4,800 psi. Which expression returns the fraction of cylinders inside that band?

- A) `=NORM.DIST(4800, 4496, 442, TRUE) - NORM.DIST(3800, 4496, 442, TRUE)`
 - B) `=NORM.DIST(4800, 4496, 442, TRUE) + NORM.DIST(3800, 4496, 442, TRUE)`
 - C) `=NORM.DIST(4800, 4496, 442, FALSE) - NORM.DIST(3800, 4496, 442, FALSE)`
 - D) `=NORM.INV(0.95, 4496, 442) - NORM.INV(0.05, 4496, 442)`
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Q2. [Matching] Match each Excel function to the quantity it returns.

Choices:

- (A) `=NORM.DIST(x,  $\mu$ ,  $\sigma$ , TRUE)`
- (B) `=NORM.INV(p,  $\mu$ ,  $\sigma$ )`
- (C) `=AVERAGE(range)`
- (D) `=NORM.DIST(x,  $\mu$ ,  $\sigma$ , FALSE)`
- (E) `=STDEV(range)`

Matches:

1. The sample standard deviation of the data
 2. The value below which the fraction p of observations falls
 3. The cumulative probability $P(X \leq x)$
 4. The arithmetic mean of the data
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Q3. [True/False] In Excel, `=NORM.DIST(x,  $\mu$ ,  $\sigma$ , FALSE)` returns the probability that X is exactly equal to x .

Q4. Concrete compressive strength measurements from your dataset have $\mu = 4,496$ psi and $\sigma = 442$ psi. Using only the 68–95–99.7 rule (no Excel), which answer best estimates $P(f_c < 4,000 \text{ psi})$?

- A) $z \approx -1.12$; estimated probability is roughly 45–50%
 - B) $z \approx -1.12$; estimated probability is roughly 10–14%
 - C) $z \approx +1.12$; estimated probability is roughly 85–90%
 - D) $z \approx -2.00$; estimated probability is roughly 2–3%
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Q5. ACI 318 requires that no more than ____% of concrete test cylinders fall below the specified compressive strength f'_c .

- A) 5%
 - B) 20%
 - C) 15%
 - D) 10%
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Q6. [Select all that apply] Which of the following are true of the Normal distribution?

- (A) It is symmetric about its mean
 - (B) About 95% of observations fall within $\pm 2\sigma$ of the mean
 - (C) It is bounded below at zero, so it cannot produce negative values
 - (D) Its mean, median, and mode all coincide
 - (E) A larger σ shifts the centre of the distribution to the right
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Q7. [Ordering] You are asked what fraction of a supplier's cylinders fall below the specified f'_c . Put the steps in the order you would carry them out.

- (A) Compute $z = (f'_c - \mu) / \sigma$
- (B) Read the cumulative probability corresponding to that z
- (C) Compute μ and σ from the cylinder test results
- (D) Compare the resulting percentage against ACI 318's 10% limit

Q8. A lognormal distribution is most appropriate when data:

- A) Is symmetric and ranges from $-\infty$ to $+\infty$
- B) Counts discrete events over a fixed interval
- C) Is left-skewed, bounded above, and whose natural log is uniformly distributed
- D) Is right-skewed, always positive, and whose natural log is Normal

Q9. The Normal distribution appears frequently in engineering measurements because:

- A) Engineers prefer simple models with only two parameters to estimate
- B) The Normal distribution fits every measured engineering quantity equally well
- C) Measured quantities are often the sum of many small independent influences
- D) Engineering design standards require that the Normal distribution be used

Q10. In your dataset, Supplier Alpha has $\sigma = 344$ psi and Supplier Beta has $\sigma = 484$ psi. Both suppliers target the same specified $f'_c = 4,000$ psi. Which statement is correct?

- A) Alpha has a higher failure rate because its σ is smaller
- B) Beta is the more consistent supplier because its σ is larger and its mean is higher
- C) Alpha is more likely to meet ACI 318's 10% limit because its σ is smaller
- D) The failure rate depends only on the mean, not σ

Q11. NORM.DIST takes a final TRUE/FALSE argument. For a **continuous** variable like concrete strength, what does each give you?

- A) TRUE gives $P(X = x)$ exactly and FALSE gives $P(X \leq x)$, the running total
 - B) TRUE returns the answer as a percentage and FALSE returns a decimal
 - C) TRUE gives $P(X \leq x)$; FALSE gives the curve's height at x , a density
 - D) They are interchangeable for a Normal distribution, which is symmetric
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Q12. With mean 4,500 psi and σ 300, which formula returns the probability that a cylinder tests **between 4,000 and 5,000 psi**?

- A) =NORM.DIST(5000, 4500, 300, TRUE) – NORM.DIST(4000, 4500, 300, TRUE)
 - B) =NORM.DIST(5000, 4500, 300, TRUE) + NORM.DIST(4000, 4500, 300, TRUE)
 - C) =NORM.DIST(1000, 4500, 300, TRUE)
 - D) =1 – NORM.DIST(5000, 4500, 300, TRUE)
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Q13. NORM.INV runs the other way — probability in, value out. You need the strength that is **exceeded by 95%** of cylinders. Which call is right?

- A) =NORM.INV(0.95, 4500, 300)
 - B) =NORM.INV(0.05, 4500, 300)
 - C) =NORM.INV(95, 4500, 300)
 - D) =1 – NORM.INV(0.95, 4500, 300)
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Q14. Cost is right-skewed, so you add a natural-log helper column with =LN(. . .). Which statement about working with the lognormal is correct?

- A) Take logs, then use the Normal functions with the mean and σ of the *logged* column
 - B) Take logs, then use the Normal functions with the mean and σ of the *original* column
 - C) LOGNORM.DIST removes the need for logs and takes the mean and σ of the raw costs
 - D) Cost data must be made symmetric by removing outliers before any of this works
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Q15. You need to compare σ between the two suppliers, with Supplier in column B and strength in column D. Excel gives you AVERAGEIF for the conditional mean but nothing equivalent for σ . Which formula gives Alpha's standard deviation alone?

- A) =STDEV(FILTER(D2:D501, B2:B501="Alpha"))
- B) =STDEVIF(B2:B501, "Alpha", D2:D501)
- C) =STDEV(D2:D501)
- D) =AVERAGEIF(B2:B501, "Alpha", D2:D501)